TOP-Q — Day 5 worked notes

Light propagation, what "the phase" means, and loss

PH 5110 term paper — Kotapati Kishore (EE26E016)

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All results are **numerical simulations** [Simulated] unless marked [Derived]. Arrays T and D as on Day 3, now with $M = 81$ guides so that no light reaches the ends before the output (largest edge power before $z = L$: 6×10^{-13}). Device length $L = 5\pi/\kappa_2 =$ **10 coupling lengths** $\pi/(2\kappa_2)$, the minimum suggested by Blanco-Redondo *et al.* Units: z in $1/\kappa_2$, b and α in κ_2.

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1 How light moves through the array

1.1 The propagator [Derived]

From $d\boldsymbol{a}/dz = iH\boldsymbol{a}$: $\boldsymbol{a}(z) = U(z)\boldsymbol{a}(0)$ with $U(z) = e^{iHz}$. Writing $H = \sum_m b_m \psi_m \psi_m^\dagger$,

$$\boldsymbol{a}(z) = \sum_m \underbrace{\langle \psi_m | \boldsymbol{a}(0) \rangle}_{\text{how much of mode } m} \; e^{ib_m z} \; \psi_m .$$

In simple words: split the input light into the array's modes; each mode just turns its phase at its own speed b_m; add them back up at the output.

Code checks [Verified-by-test]: U is unitary and $U(z_1)U(z_2) = U(z_1 + z_2)$; total power stays 1 to 10^{-12}; results agree with an independent ODE solver (SciPy `solve_ivp`, 8th-order Runge–Kutta) to 10^{-8}, with and without loss.

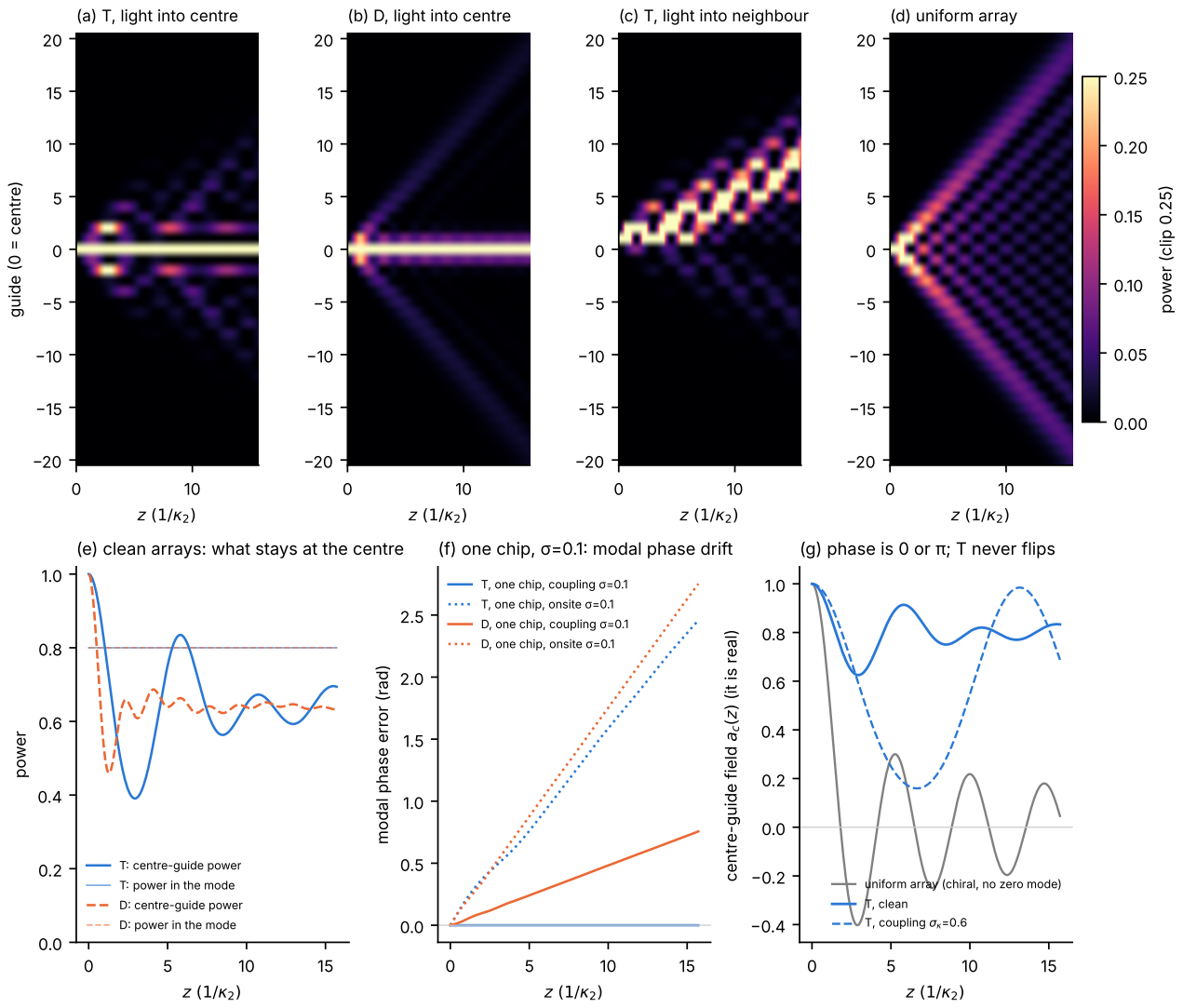


Figure 1: **Simulation**, generated by `scripts/day5_propagation.py` from `data/day5_propagation.npz`. (a)–(d) Power in each guide vs. z (top views, like a fluorescence image). (e) Power left in the centre guide and power in the defect mode. (f) Phase error of the defect-mode amplitude on one disordered chip ($\sigma = 0.1$). (g) The centre-guide field: always real; only T never changes sign.

1.2 Launching into the centre guide

Putting all light into the centre guide c sends a fraction $|\psi_0(c)|^2$ into the defect mode. For both T and D this is $(1 - r^2)/(1 + r^2) = 0.8$ [Derived; Verified-by-test] — the IPR matching of Day 3 also makes the launch efficiency equal. The other 20% goes into bulk modes and spreads out (Fig. 1a,b). Fig. 1c: light put into the guide *next to* the T centre does not excite the zero mode at all (the zero mode is exactly zero there) and walks away. Fig. 1d: a uniform array shows pure **discrete diffraction**, two outgoing lobes — the same pattern Blanco-Redondo *et al.* measured for their equidistant reference array [Literature].

2 What do we mean by "the phase"?

There are (at least) two honest definitions, and they answer different questions:

	Modal phase $\varphi_{\mathrm{m}} = \arg\langle\psi_0\|\boldsymbol{a}(L)\rangle$	**Output-guide phase** $\varphi_{\mathrm{g}} = \arg a_c(L)$
What it is	phase of the light that is in the defect mode	phase of the light in the centre output guide
Who sees it	a perfect mode filter	an interferometer that taps that one guide
Clean value	$b_0 L$ (T: 0; D: $\frac{20}{9} L$)	$b_0 L$ + small wiggles from bulk light

Both are measured **relative to** $\beta_0 L$. The common term $\beta_0 L$ is the same for every guide and cancels when two arms share it. The phase *error* we care about is (disordered chip) − (clean chip) at $z = L$. To first order, $\delta\varphi_{\rm m} \approx \delta b\, L$: **the phase error grows in proportion to the length**. Fig. 1f, one chip with on-site $\sigma_\beta = 0.1$: T drifts by 2.4 rad and D by 2.7 rad over 10 coupling lengths. With coupling disorder, T does not drift at all; D drifts by 0.76 rad.

3 A new result: chiral symmetry makes the phase 0 or π

3.1 The theorem [Derived; Verified-by-test]

Launch in guide c and look at the same guide. Then

$$a_c(z) = \langle c|e^{iHz}|c\rangle = \sum_m |\psi_m(c)|^2 e^{ib_m z}.$$

If H is chiral, the modes come in pairs ψ_m and $\Gamma\psi_m$ with $b \to -b$, and $|\Gamma\psi_m(c)| = |\psi_m(c)|$. Each pair adds $|\psi_m(c)|^2(e^{ibz} + e^{-ibz}) = 2|\psi_m(c)|^2 \cos bz$, which is *real*. So

$$\boxed{a_c(z) \text{ is real for every } z \text{ in ANY chiral array } \Rightarrow\ \varphi_{\rm g} \in \{0, \pi\}.}$$

Numerically $|\mathrm{Im}\, a_c| < 3 \times 10^{-15}$ for the uniform array, clean T, and T with $\sigma_\kappa = 0.6$.

3.2 What topology adds: no sign flip

The zero mode contributes a constant $w_0 = |\psi_0(c)|^2$, and all the rest together has size at most $1 - w_0$, so

$$\mathrm{Re}\, a_c(z) \ \geq\ 2w_0 - 1.$$

If $w_0 > 1/2$, the field can never cross zero, so the phase stays at 0 at *every* length. Clean T: $w_0 = 0.8$, so $a_c \geq 0.6$ (measured minimum 0.62). The uniform array is also chiral but has no zero mode: its field swings negative (minimum −0.40), so its phase jumps between 0 and π (Fig. 1g).

This is the cleanest answer so far to the research question. For a chiral array, *chiral symmetry alone* (not topology) forces the self-phase to be 0 or π. *Topology* (the zero mode) is what keeps it at 0 rather than flipping, *as long as* the zero mode keeps more than half of its weight on the input guide. Day 4 showed this weight falls to 0.61 at $\sigma_\kappa = 0.6$ and 0.54 at 0.8 — close to the 0.5 limit. And all of this fails the moment chiral symmetry is broken (on-site errors).

4 Loss

4.1 Uniform loss [Derived; Verified-by-test]

Power loss α per unit length: $H \to H + i\frac{\alpha}{2}\mathbb{1}$. Because $\mathbb{1}$ commutes with H, $\boldsymbol{a}(z) = e^{-\alpha z/2} \times$(lossless result): **power drops as $e^{-\alpha z}$ and every phase is unchanged** (measured change $\leq 2 \times 10^{-14}$ rad, Fig. 2a).

4.2 Random loss: the zero mode's phase is still pinned [Derived; Verified-by-test]

Let each guide have its own loss: $H = H_0 + iD$, with D real and diagonal and H_0 chiral. Then

$$\Gamma H \Gamma = -H_0 + iD = -(H_0 - iD) = -H^\dagger.$$

This forces the eigenvalues into pairs $(b, -b^*)$. A single isolated mode must be its own partner: $b = -b^*$, i.e. $\mathrm{Re}\, b = 0$ *exactly*. Its *decay* ($\mathrm{Im}\, b$) changes, but its *phase velocity* does not. Numbers (500 chips per point, loss uniformly random in $[0, 2\bar{\alpha}]$): T: std of $\mathrm{Re}\,\delta b \approx 10^{-15}$ for all $\bar{\alpha}$; D: 10^{-6} to 10^{-3} (a second-order effect). Both lose amplitude at the same average rate $\approx \bar{\alpha}/2$ (Fig. 2b). [Literature: the idea of a zero mode surviving in a lossy SSH lattice was demonstrated in waveguides by Weimann *et al.*, Nature Materials **16**, 433 (2017), doi:10.1038/nmat4811 — read it before citing for this exact statement.]

In simple words: loss makes the light dimmer. Uniform loss dims everything equally and changes no phases. Random loss also makes the topological mode dimmer by a random amount, but it still does not change its phase. So for T, loss turns into an *amplitude* error, not a phase error. For an interferometer, an amplitude error between the two arms still lowers the contrast (Day 8).

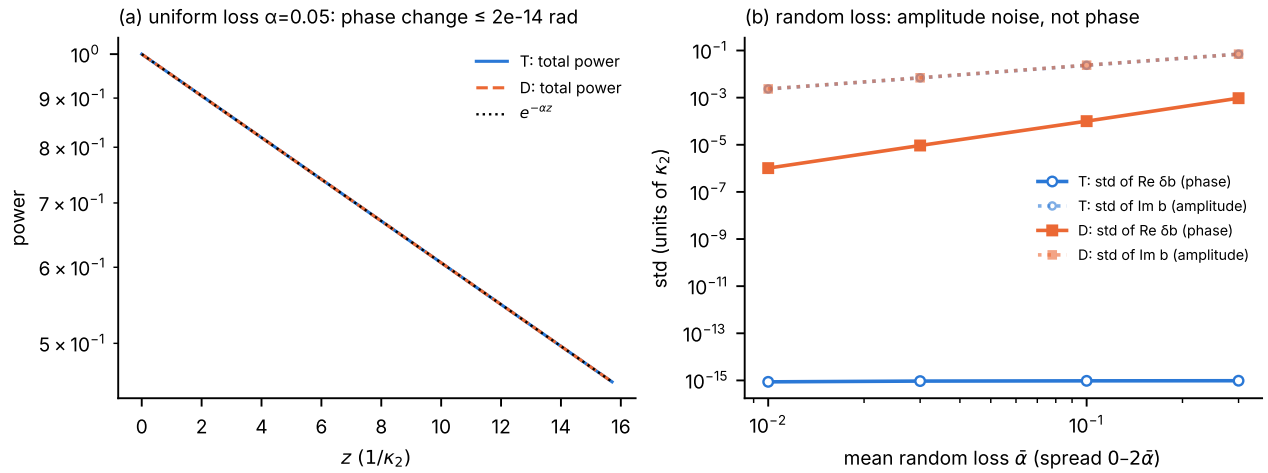


Figure 2: **Simulation** (`data/day5_propagation.npz`). (a) Uniform loss: pure exponential, no phase change. (b) Random loss: spread of the phase rate $\mathrm{Re}\,\delta b$ (solid) and of the decay rate $\mathrm{Im}\,b$ (dotted). T's phase rate is pinned at round-off level.

5 Five different things — now with evidence

Property	Protected by	Broken by
Eigenvalue pinning ($b = 0$)	chiral symmetry (exact)	on-site / width errors
Localisation at the input	topology + gap; weakens with strong disorder	strong coupling disorder (Day 4)
Self-phase $\in \{0, \pi\}$	**chiral symmetry alone**	on-site / width errors
Self-phase $= 0$ (no flip)	topology: zero-mode weight $w_0 > 1/2$	$w_0 < 1/2$; on-site errors
Phase rate under random loss	non-Hermitian chiral pairing $(b, -b^*)$	on-site errors
Transmitted power	nothing special: same mode loss as D	loss of any kind

6 Hardware: how propagation and loss are measured in real labs

We did none of these measurements. The images shown in our chat (from published papers) illustrate the real methods:

- **Top-view imaging of light inside an array.** In femtosecond-laser-written glass waveguides, light scattered or fluoresced from the waveguides is filmed from above; the pictures look like our Fig. 1a–d. Silicon chips cannot be viewed this way easily, so only the output is seen.
- **Output-facet imaging.** The light leaving the chip edge is focused onto an infrared camera; each bright spot is one waveguide. This gives $|a_n(L)|^2$ but *not* the phase. To get a phase, the output must interfere with a reference beam — i.e. an interferometer, which is exactly why Days 8–10 build MZIs.
- **Cut-back loss measurement.** Measure the transmitted power for several waveguide lengths and fit a straight line on a log scale; the slope is α. This is the experimental version of Fig. 2a.

Look up and cite the source papers shown with each image before using them ("Reproduced from [ref]").

7 Code and tests

New: `topq/propagation.py` (loss, propagator, propagate, launch, modal amplitude, guide field), `scripts/day5_propagation.py`, `tests/test_propagation.py` (15 tests, including the real-amplitude theorem, the $2w_0 - 1$ bound and the loss pinning). Suite: **93 passed**. Run time of the script: about 30 s.

8 Self-test

1. Why does light launched next to the T centre not excite the zero mode?

2. Prove in three lines that $a_c(z)$ is real in a chiral array.
3. Why does the uniform array's centre field change sign but T's does not?
4. Why does uniform loss change no phase?
5. Show that $\Gamma(H_0 + iD)\Gamma = -(H_0 + iD)^\dagger$ forces $\operatorname{Re} b = 0$ for an isolated mode.
6. Why does a longer device make phase errors worse?